Environmental drivers of low vaccine responsiveness in a lab-to-wild rodent model

Simon A. Babayan[1,†,*], Saudamini Venkatesan[2,†], Jessica Hall[1,2], Ewan Smith[1], Amy Sweeny[2], and Amy B. Pedersen[2,*]

[1]School of Biodiversity, One Health & Veterinary Medicine, University of Glasgow, Glasgow, UK
[2]Institute of Ecology and Evolution, School of Biological Sciences, University of Edinburgh, Edinburgh, UK
[†]These authors contributed equally to this work
[*]Corresponding authors: simon.babayan@glasgow.ac.uk, amy.pedersen@ed.ac.uk

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ABSTRACT

Vaccination is the most effective medical intervention to prevent infectious disease and protect public health. However, most new vaccines fail late in the clinical trial pipeline, and even established vaccines often perform worse in populations that differ substantially from those in which they were initially tested. This can lead to breakthrough infections and prevent or delay herd immunity. For anthelmintic vaccines, the situation is worse still: none have progressed beyond clinical trials in humans to date. Despite decades of immunological research and vaccine development, the causes of vaccine hyporesponsiveness remain difficult to identify, quantify, and therefore, address. We and others have shown that diet could have profound impacts on anthelmintic immunity: while protein restriction disproportionally impairs type 2 immunity compared to pro-inflammatory and type 1 immunity, dietary supplementation with a nutrient-rich chow can help wild rodents reduce parasite burdens by $\sim$ 50%. Here, we therefore hypothesised that the leading causes of type-2 vaccine hyporesponsiveness are environmental, and are exacerbated when individuals are under nutritional or physiological (e.g., reproductive) stress. We used paired cohorts of laboratory-reared and wild wood mice (*Apodemus sylvaticus*) vaccinated with a single or two doses of diphtheria toxoid vaccine formulated with alum to drive type 2 responses. We show that circulating vaccine-specific IgG1 antibodies dropped on average by 50% in a wild population compared to individuals of the same species that had been reared under controlled laboratory conditions. We also found that, regardless of habitat, substantial variation in vaccine responsiveness was explained by diet, alongside weaker negative effects of being male and reproductively active. Surprisingly, helminth infection did not directly suppress vaccine-specific antibody production once habitat, diet, and sex were accounted for. While much variation within the habitat remains to be explained, our results indicate that the environment, chiefly dietary resource, plays a dramatic role in vaccine responsiveness. Further, they suggest that laboratory settings may systematically overestimate vaccine performance by failing to capture natural environmental variability. We anticipate that our study will inaugurate a robust modelling approach and biological system to disentangle and quantify the complex factors that cause vaccine responsiveness in target populations. It may also call for a need to systematically raise efficacy thresholds for initiating clinical phases of vaccine development.

INTRODUCTION

Unanticipated variability in individual responses to vaccination can lead to a lack of protection from the target disease, continued pathogen transmission, and an increased risk of the evolution of vaccine escape[1–4]. Such heterogeneity in vaccine efficacy is largely driven by the individual's baseline immune state and responsiveness to novel antigen, affecting both innate and adaptive immunity, thereby potentially driving disease and transmission[5], and are themselves the results of earlier interactions between intrinsic and environmental factors, including genetics[6] and epigenetics[7], sex[8], reproductive status[9], prior infections[10–13], and diet[14–16]. Such heterogeneities can have stark implications for human health, manifesting for example in the substantial drop in vaccine immunogenicity and efficacy in low-income or rural relative to high-income or urban populations[17,18]. These impediments to effective vaccination are exacerbated for vaccines against helminths, for which the most at-risk individuals are found in rural and low-income populations[19,20]. Furthermore, helminths are reported to cause poor vaccine responsiveness due to their ability to suppress their host's immunity[21,22], potentially compounding effects of environment and immune regulation.

Unsurprisingly, only a small minority of vaccines successfully transition from the preclinical stages of development through phase III trials and reach mass market[23], and there are still no antihelmintic vaccines for human helminthiases. Furthermore, even the most successful vaccines can suffer substantial loss of efficacy, notably for people of different ancestry[24] or different habitats[18] than the populations in which they are initially deployed. For example, individuals of Tanzanian ancestry were reported to have higher inflammatory and type 1 immune markers[25] and responses to measles vaccination[26] than those of European ancestry, while in rural populations the expression of BCG- and tetanus-specific cytokine recall and antibody responses was found to be markedly lower than in urban populations, while humoral responses to tetanus vaccination were higher[27,28].

Similar patterns have been observed across multiple vaccines[18]. Taken together, these reports indicate that vaccine efficacy depends on both the vaccine and the recipients' characteristics, and that there are substantial gaps in our ability to predict and address lack or loss of vaccine efficacy across individuals and populations. Indeed, given the complex processes that regulate protective immune responses, it is unclear how all these factors interact with each-other and with the environment, and which among them could be intervened upon to maximise vaccination efficacy.

To increase the generalisability of experimental models of immunology and vaccinology, recent efforts have been made to integrate natural variability into immunological research[29–33]. These include exposing mice to richer, "dirtier" environments, littermates, and microbiota, which stimulates mice to develop more mature, and thus more representative, immune systems and responses to infection. However, there is still uncertainty about how this heterogeneity affects the innate and adaptive arms of the immune system[34], and what consequences this might have for vaccine responsiveness. More specifically, it is unclear how environmental factors and past stressors influence organisms' responses to immunisation. This complexity obscures the causal relationships between environmental and immunological processes, making the design and interpretation of *in vivo* vaccination experiments difficult and their impact on their intended target populations less tractable.

To disentangle the relative contributions of environmental variables to vaccine responsiveness, we conducted randomised controlled vaccination and dietary supplementation in a coupled laboratory-wild rodent study system, the wood mouse *Apodemus sylvaticus*. Using the same assays and interventions in the same genetically outbred species across habitats allowed us to better isolate the effect of the wild habitat by minimising confounding effects of species as well as reagent availability and specificity. Although supplementation of wild populations prevents accurate measure of individual dietary intake and feeding preference (wild mice are free to forage), we have previously shown in the same wild populations that diet supplementation drives a sustained reduction in helminth (*Heligmosomoides polygyrus*) burdens and transmission, while also improving the efficacy of anthelmintic treatment[35]. We therefore predicted that type 2 anthelmintic vaccine responses in wild mice would be lower than in lab wood mice and higher in diet-supplemented than normal-diet animals. To test these predictions we constructed a structural causal model (SCM)[36] to partition the causal effects of habitat, diet, sex, and parasite infection on vaccine-specific IgG1 responsiveness and to identify the paths through which their effects are mediated. This causal framework, which uses graphical models to inform adjustment strategies[37,38], can eliminate or reduce multiple sources of confounding, such as common causes (misattribution of causal effects), sampling biases (non-representative test population), and transportability biases (lack of generalisability). We used Bayesian models to estimate the average direct and indirect causes of loss of vaccine efficacy when translating a vaccine from the laboratory to the wild, and to accurately model how uncertainty and biological variability propagate through the causal structure.

RESULTS

Vaccination induced stronger but variable antibody responses in the laboratory than in the wild

Wild and laboratory *A. sylvaticus* wood mice were given a high- or low-quality diet and then immunised with a diphtheria-alum vaccine (DTV) once or twice. DTV-specific IgG1 serum concentrations were then measured by ELISA seven to 35 days post immunisation (see details in Materials, Methods, and Models; Figure 1).

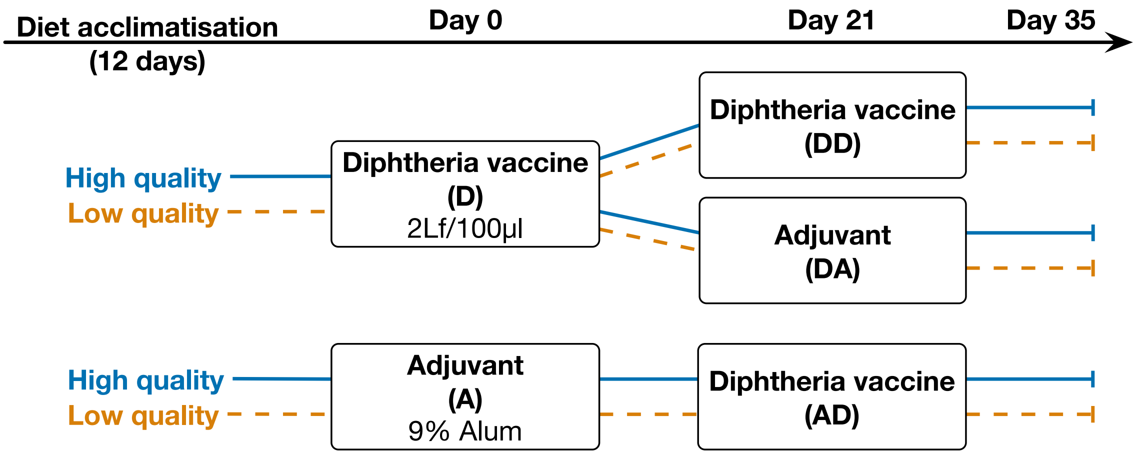


Fig. 1: Allocation of diet and vaccines. Wood mice from a laboratory colony, and from the wild were provided enriched (TransBreed) or standard (lab: normal chow; wild: no supplementation) diets. After at least 12 days, mice were allocated at random to immunisation with a 2Lf/100µl Diphtheria Toxoid + 9% alum vaccine (D) or with adjuvant alone (A). Twenty-one days after their first immunisation, mice that had been immunised with D were then given a second shot of the vaccine (DD) or adjuvant alone (DA), while mice that had received an initial adjuvant vaccine (A) were all given D for their second shot (AD).

Wild mice generated 46.9% $\pm$ 0.1 % lower DTV-specific IgG1 antibody responses than their lab-based conspecifics, regardless of the vaccine regimen or diet (Figure 2A, 2B 'Hab'). While immunisation was most effective with two doses in both habitats, lab mice reached peak antibody levels with only a single dose (Figure 2A, 'DA' vs 'DD'; Lab boosted vs. Lab non-boosted OD = 0.2 $\pm$ 0.1 whereas Wild boosted vs. Wild non-boosted OD = 1.02 $\pm$ 0.23). However, in the wild, only when given two doses did the mice show responsiveness similar to the level observed in laboratory mice (Figure 2A, 'DD'; difference between Wild non-boosted vs. Lab non-boosted OD = 0.75 $\pm$ 0.11 while difference between Wild boosted vs. Lab boosted OD = 0.61 $\pm$ 0.18; LRT of Habitat $\times$vaccine regimen: $\Delta dof = 4, \chi^2 = 30.6, p < 0.001$). Only mice that were administered the DT antigen generated DTV-specific IgG1, indicating that there was no pre-existing immunity to this antigen in our study populations (Figure 2A, group 'A').

To further quantify how habitat (lab *vs.* wild), diet (supplemented *vs.* control), and the immunisation protocol were associated with variation in vaccine responsiveness, we constructed a multilevel Bayesian model of those factors with varying intercepts for each

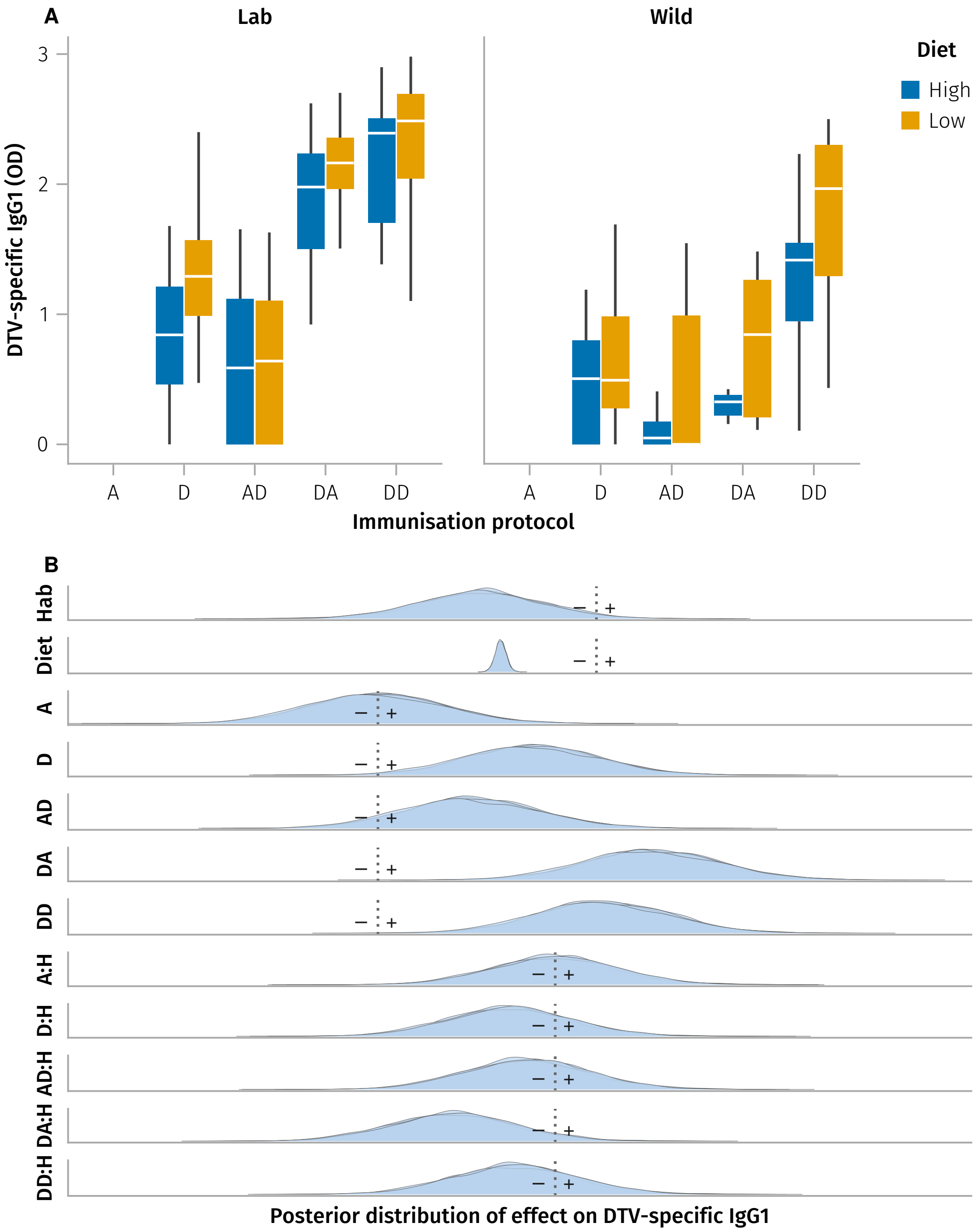


Fig. 2: Vaccine-induced DTV-specific IgG1. A, diphtheria toxoid vaccine (DTV)-specific IgG1 optical density in mice from laboratory- and wild-habitats provided high and low quality dietary supplements. Mice were immunised either once, with adjuvant as control (A) or DTV (D), or twice, using adjuvant alone followed by DTV (AD), DTV and subsequently adjuvant alone (DA), or two separate doses of DTV (DD). Only mice immunised at least 7 days prior to serum ELISA sampling are shown. **B**, posterior conditional distributions of the coefficients for diet, habitat, and vaccine formulation (A, D, AD, DA, DD), and effect of the same vaccines. Each density curve represents the distribution of 3,000 samples from one of four MCMC chains; dotted line represents the the value for which the reference (global intercept for Diet and Habitat; Adjuvant alone for the vaccines) is null (see parameter estimates in Table 1).

immunisation protocol (Figure 2B, Table 1). This model indicated that the effects of each vaccine formulation on DTV-specific IgG1 OD were substantially more variable than those of habitat and diet. Further, it shows that vaccine responsiveness for all regimens but especially DA, i.e. a single early immunisation, was substantially degraded in the wild, indicating a faster loss of vaccine-induced immunity in the wild than in the lab (Figure 2B, Table 1).

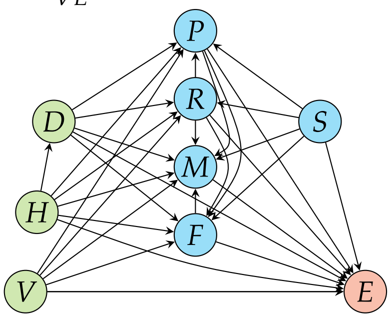
However, this model does not explain how habitat and diet are driving this loss of efficacy.Therefore, it does not indicate which mechanisms might be targeted to prevent this loss of responsiveness. Further, it does not account for potential confounding and mediating effects of sex, reproductive status, body mass, body fat, or parasite burdens, which we hypothesise to contribute to variation in vaccine responsiveness. To address these limitations, we then constructed a structural causal model (SCM) of the processes generating variation in vaccine responsiveness $\mathbb{C}_{VE}$ in this study system.

Construction of a structural causal model of vaccine responsiveness

To understand more specifically the relative contributions of diet and habitat to vaccine responsiveness, and how that effect might be mediated by parasite infection, reproductive status, and body condition, we built a causal model of vaccine efficacy in the wood mouse system, first by constructing a graphical model of our hypotheses, then using our data to validate and refine its structure (see casual flow diagram, Fig. S1).

The resulting causal directed acyclic graph (DAG) $\mathbb{G}_{VE}$ and corresponding SCM $\mathbb{C}_{VE}$ then allowed us to estimate vaccine responsiveness E as a function of vaccination V, habitat H and diet supplementation D, and more importantly, to identify which variables could act as confounders or mediators that might modulate vaccine responsiveness. In summary, the DAG $\mathbb{G}_{VE}$ (Figure 3A) represents the hypothesised structure of the biological processes that cause the variation in vaccine responsiveness observed in this system, and the corresponding SCM $\mathbb{C}_{VE}$ (Figure 3B) that links (via the edges in $\mathbb{G}_{VE}$) our variables

A $\mathbb{G}_{VE}$:



B $\mathbb{C}_{VE}$:

$$
\begin{aligned}
V &:= \mathcal{U}_V \\
H &:= \mathcal{U}_H \\
S &:= \mathcal{U}_S \\
D &:= \beta_{HD}H + \mathcal{U}_D \\
R &:= \beta_{DR}D + \beta_{HR}H + \beta_{VR}V + \beta_{SR}S + \mathcal{U}_R \\
P &:= \beta_{TP}V + \beta_{DP}D + \beta_{HP}H + \beta_{RP}R + \beta_{SP}S + \mathcal{U}_P \\
F &:= \beta_{TF}V + \beta_{DF}D + \beta_{HF}H + \beta_{SF}S + \beta_{RF}R + \beta_{PF}P + \mathcal{U}_F \\
M &:= \beta_{VM}V + \beta_{HM}H + \beta_{SM}S + \beta_{RM}R + \beta_{PM}P + \beta_{FM}F + \\
&\quad \beta_{DM}D + \mathcal{U}_M \\
E &:= \beta_{VE}V + \beta_{HE}H + \beta_{FE}F + \beta_{DE}D + \beta_{ME}M + \beta_{RE}R + \\
&\quad \beta_{PE}P + \beta_{SE}S + \mathcal{U}_E
\end{aligned} \tag{1}
$$

Fig. 3: Causal Model of vaccine responsiveness. A, Directed acyclic graph $\mathbb{G}_{VE}$ representing hypothesised causal effects driving variation in efficacy E of vaccine V in laboratory or wild mice H, under supplemented or control diet D (green nodes depict experimental treatments). Covariates, mediators, and potential confounders (blue nodes) were time since vaccination T, parasite infection (presence or burden) P, reproductive status R, body mass M, fat scores F, and sex S. Directed edges depict causal effects hypothesised to operate between nodes. **B**, Structural causal model $\mathbb{C}_{VE}$ derived from $\mathbb{G}_{VE}$ with coefficients β_{ij} representing the effect of cause i on effect j. $\mathcal{U}_j$ represents the unmeasured variation of the response variable, not represented in $\mathbb{G}_{VE}$ (panel A) for clarity.

of interest (nodes in $\mathbb{G}_{VE}$) to vaccine efficacy E under the different habitat and dietary treatments.

- Vaccination V was allocated at random, and therefore has no parent node. It was hypothesised to potentially affect parasite infection P, reproductive status R, body mass M and fat stores F through non-specific effects in addition to its main intended effect on vaccine responsiveness E.

- Whether a mouse was born in the lab or in the wild habitat H was considered random and therefore took no parent node, but was hypothesised to potentially affect all other variables except V and sex S.

- Diet D was allocated at random in the laboratory colony, but because wild mice could also forage freely, we considered that being born in the wild could have a causal effect on diet, specifically on the effect of supplementation relative to the control laboratory or the wild diet. We therefore considered a causal effect of H on D. Diet was hypothesised to affect reproductive status R, body mass M, body fat F

and parasite burdens P as well as E, with fat scores F hypothesised to be a cause of variation in body mass M.

- Sex S being genetically determined and therefore subject to Mendelian randomisation[39], and all treatments being randomised with respect to sex, S had no parent node, and was considered causally prior to reproductive status R, body mass M, body fat F and parasite burdens P.
- Reproductive status R being driven by sex, it was hypothesised to affect body mass M and parasite burden P.
- Vaccine responsiveness E was hypothesised to be affected by all other variables.
- Each node also had its own error term to capture all unobserved sources of variation, assumed to be independent and identically distributed (Figure 3B).

Validation of the causal model

We then tested the validity of the causal hypotheses described above, using $\mathbb{G}_{VE}$ to construct each model. For example, $\mathbb{G}_{VE}$ (Figure 3A) indicates that parasite infection should be independent of vaccination if we adjust for diet, reproductive status, sex, and habitat ($P \perp\!\!\!\perp V \mid D, R, S, H$); therefore, we fitted a binomial generalised linear mixed model $P \sim \alpha + \beta_V V + \beta_D D + \beta_R R + \beta_S S + \beta_H H + (1|ID)$, and confirmed that the posterior distribution of β_D was not significantly different from zero. This process was repeated for all implied independencies derived from the causal graph $\mathbb{G}_{VE}$ (see Model validation methods), and $\mathbb{G}_{VE}$ was updated until all causal paths were both supported by the observed data and consistent with biological expectations.

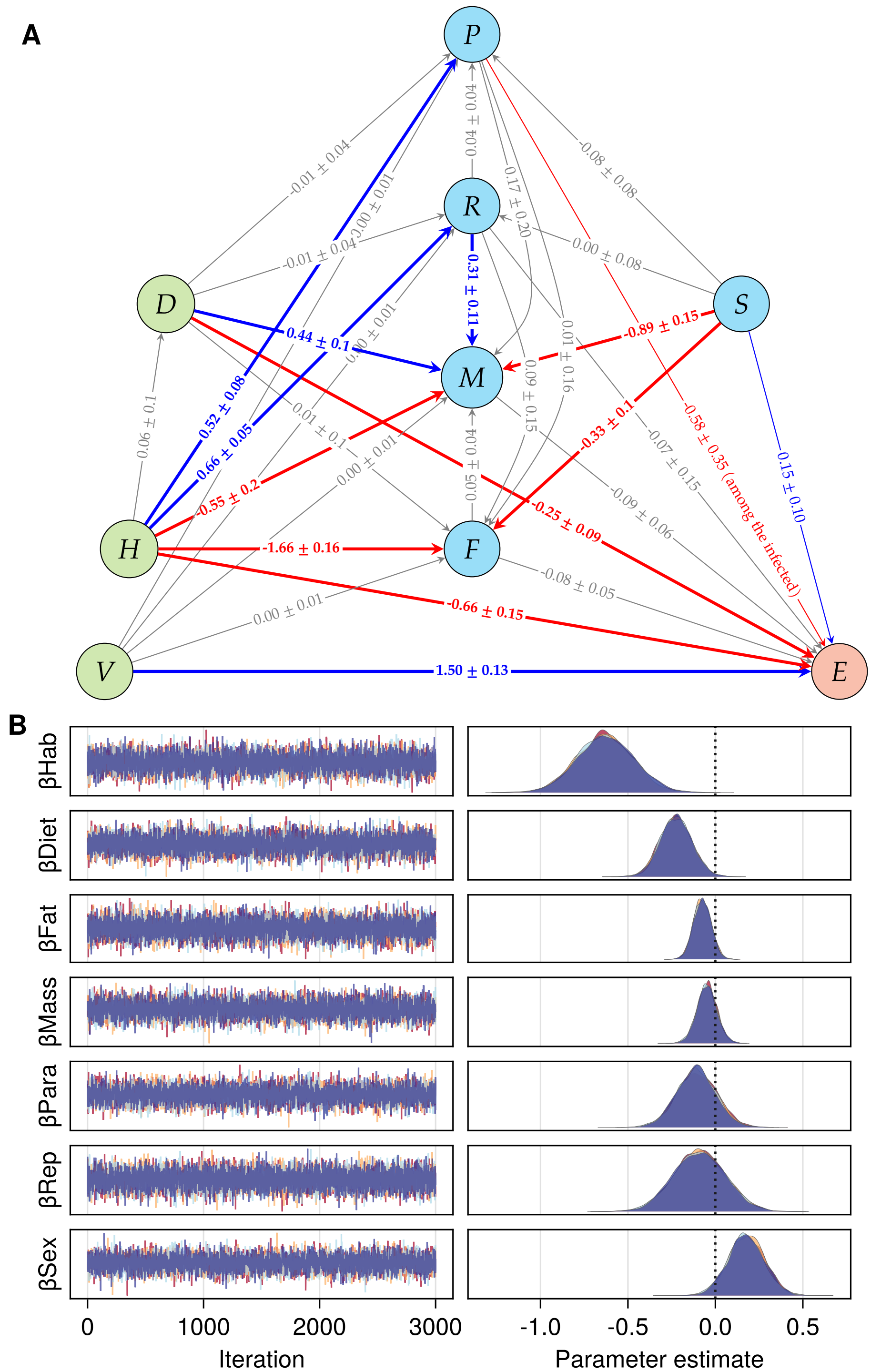


Fig. 4: Fully resolved Causal Model of vaccine efficacy. **A**, Directed acyclic graph representing point estimates of positive (blue) and negative (red) direct causal effects ± std driving vaccine responsiveness E. **B**, MCMC sampling of posterior distributions of the coefficients estimating the direct effect of the wild habitat H on vaccine responsiveness E, with the adjustment set including, Diet D, Fat scores F, body mass M, parasite burden P, active reproductive status R, and female sex S; vaccine formulation V and individual treated as random effects: $E := \beta_{Hab}H + \beta_{Diet}D + \beta_{Fat}F + \beta_{Mass}M + \beta_{Para}P + \beta_{Rep}R + \beta_{Sex}S + \mathcal{N}(0, Exp(\sigma_E))$ (see Model parameterisation methods). Note that each edge in panel A shows the direct causal effect of its parent node on its child node, which is conditional on the specific adjustment set for that model; thus, only the coefficient β_{Diet} for the edge between D and E is a causal estimate in panel B, while the others are only associative.

Causal analysis indicates preponderant effects of diet and habitat on vaccine responsiveness

Using the resulting, validated, DAG, we parametrised $\mathbb{C}_{VE}$ (Model 1) using Bayesian linear models to estimate the total and direct causal effects operating within $\mathbb{G}_{VE}$ (Fig. 4). Vaccination had an average causal effect of 1.50 ± 0.13 ($log10 + 1$)-transformed OD units ($logOD$) relative to unvaccinated animals and no indirect effects mediated by body fat F, body mass M, reproductive status R or parasite burden P (Fig. 4B). However, vaccine responsiveness was strongly influenced by both wild habitat and dietary supplementation.

Habitat had wide-ranging effects on wood mice and their response to vaccination

Living in the wild had a total negative effect on vaccine-specific IgG1 of -0.46 $\pm$ 0.24 *logOD*, partitioned between body mass M with a direct negative impact of -0.66 $\pm$ 0.15 *logOD* (corresponding to wild wood mice being on average 1.3 $\pm$ 0.2 g lighter than their laboratory counterparts), and body fat F (reduction by 2.3 fat score points on the 6-point scale).

Wood mice in the wild but not in the lab could freely reproduce and were naturally exposed to *H. polygyrus*, reflected in direct causal effects of 0.66 $\pm$ 0.05 and 0.52 $\pm$ 0.08 of H on R and P, respectively. Consistent with expectations under random diet allocation, there was no effect of habitat H on diet D.

Diet supplementation reduced vaccine responsiveness

The total causal effect of diet supplementation on vaccine responsiveness was estimated to be -0.25 $\pm$ 0.09 *logOD*, adjusted for habitat. However, the direct effect of diet supplementation on vaccine responsiveness was estimated to be -0.23 $\pm$ 0.09 *logOD*, when adjusting for body mass, fat stores, habitat, sex, reproductive status, parasite burden, and vaccine formulation. This indicates that the effect of diet supplementation on vaccine responsiveness was mostly direct (or mediated by unobserved variables along the path $D \rightarrow E$, e.g. immune pathways affecting DTV-specific IgG1 production or the microbiome), but was not mediated by its effects on body mass, fat stores, reproductive status, or parasite burden. Dietary supplementation increased body mass on average by 2.53 $\pm$ 0.3 g, though did not affect body fat scores.

Sex had both direct and indirect effects on vaccine responsiveness

The total causal effect of sex on DTV-specific antibody production E indicated greater vaccine responsiveness in females (0.26 ± 0.15 $logOD$). However, only a little more than half of the effect of sex (0.15 ± 0.10 $logOD$) was direct, with body mass (0.08 ± 0.01 $logOD$) and body fat (0.03 ± 0.005 $logOD$) mediating the rest (Fig. 4).

Negative effect of parasite burdens on vaccine responsiveness among the infected

When considering the entire population, including uninfected mice, *H. polygyrus* burden had no clear effect on DT vaccine responsiveness (-0.13 ± 0.11 $logOD$). However, because parasite infections are overdispersed and zero-inflated, we also estimated the causal effects of infection burden on DT vaccine responsiveness only among infected wild wood mice. Parasite burden among the infected had a negative direct causal effect on vaccine responsiveness (-0.58 ± 0.35 $logOD$), when adjusting for diet, sex, reproductive status, and vaccination.

OPTIONAL: Prediction of the effect of population-wide anthelmintic treatment on vaccine responsiveness

Using the fully-parametrised SCM, we can make predictions of the effects of population-wide anthelmintic treatment on vaccine responsiveness.

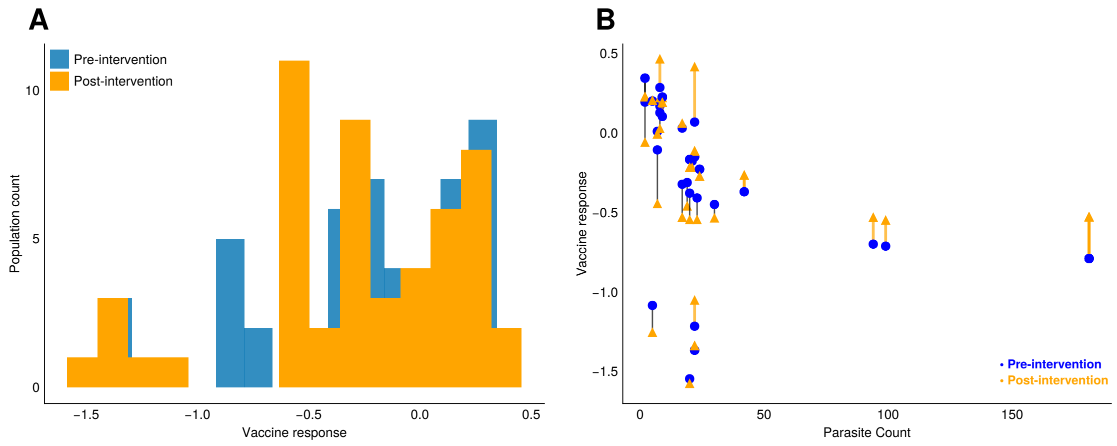


Fig. 5: Effect of simulated population-wide anthelmintic intervention on vaccine responsiveness. **A**, Effect on population-wide vaccine responsiveness. **B**, Effect on individual vaccine responsiveness as a function of pre-intervention parasite burden.

DISCUSSION

We built a structural causal model of vaccine immunogenicity to estimate how the effects of living in a natural wild habitat and of diet quality were partitioned among parasitological and physiological traits at individual and populations levels. Our three main observations were that there was substantial variation in vaccine efficacy between individuals, that both the wild habitat and dietary supplementation resulted in lower vaccine responsiveness, and that the (negative) effect of parasites on vaccine responsiveness was only detectable among the infected, but not at the whole population level. This emphasises the importance of accounting for the interplay between environmental and individual factors when analysing and predicting responses to immunisation, consistent with reports across multiple habitats and species, including humans[27,40–42].

Indeed, there was a marked hyporesponsiveness in wild wood mouse populations relative to their laboratory conspecifics immunised with the same Diphtheria Toxoid vaccine formulations. The detrimental effects of the natural habitat on vaccine responsiveness were only partially rescued by a booster dose of the vaccine. This reduction is consistent with widespread reports of vaccine failure during clinical trials, notably in phase III where vaccines are tested for efficacy in naturally exposed individuals[23], or in already-established vaccines in rural populations[18,43]. Part of the loss of vaccine responsiveness in the wild habitat was mediated by the gastrointestinal parasite *Heligmosomoides polygyrus*, consistent with reports in humans and animals showing that parasitic helminths can negatively impact vaccination[10,44–47]. These reports are not unanimous, however, with many reporting no[48] or even a positive association between vaccine responsiveness and parasitic infection[11,12]. These discrepancies might be explained by the vaccine formulations and by the immune measurements chosen. However, our present study suggests that conflicting inferences might be due to by how the effects of parasite infection are estimated: we only detected a negative effect of parasite burden among the infected animals, and no effect when uninfected individuals were included in the analysis. Thus, we might predict that parasites, and their removal prior to vaccination, would cause detectable population-wide effects on vaccination only where prevalence and burdens are very high. The mechanisms by which parasites affect vaccine responsiveness were not evaluated. Typically, two hypotheses have been proposed: counter-regulation between different arms of the immune system, e.g., Th2 responses suppressing innate and type 1 responses that drive vaccine-specific antibody production[49,50]; and immune suppression by the parasites including viruses. Animal studies have highlighted the immunoregulatory effects of gastrointestinal parasites, in particular of *H. bakeri*[22,51]. Here, the immune system of wood mice naturally infected by *H. polygyrus*, which is distinct from the laboratory model[52], might similarly be actively suppressed by the parasite. Alternatively, animals inherently more susceptible to helminth infection might have a generally lower propensity to generate strong antibody responses, which we detect "collaterally" as poor anti-vaccine responsiveness. Resolving these alternatives would require testing the prediction of both vaccine responsiveness and helminth burden from immune profiling before both infection and vaccination (see for example[53,54]). While our study focussed on helminths, other infections are likely to have impacted vaccine responsiveness in the wild, including viruses, bacteria, and other helminths. In particular, reports that human populations' variability in responses to SARS-CoV2 could be explained by cytomegalovirus infections[13], and given the 10–70% prevalence of gammaherpesvirus infections in wood mouse populations[55], it is likely that this virus could have impacted vaccine responsiveness in our study.

The dietary supplementation generated unexpected immunological effects. First, diet had no detectable effect on parasite burdens, but caused a weaker response to vaccination. The lack of detectable differences between control and supplemented grids diverges from our previous studies, where the same diet, in the same populations of the same species, reduced parasite burdens by approximately half[35]. This might be due to more abundant natural food availability in our control grids than in earlier studies, eliminating differences in antihelminthic immunity. The second unexpected outcome was that dietary supplementation caused a loss of vaccine responsiveness. This was surprising, as based on our earlier work showing increased immunity to parasite infection[35], we had predicted that diet supplementation would increase vaccine responsiveness. Taken together, (a) the positive or null effect of diet on antiparasitic immunity and (b) its detrimental effect on vaccine responsiveness, indicate that nutrition differently affects immune pathways underlying vaccine responses and antihelminthic immunity. In addition, because immune responses were not uniformly reduced under lower diet availability whether in the laboratory colony or in the wild, we are not observing starvation or generalised impairment of the immune system. Rather, we hypothesise that the immune system adopts different strategies depending on the availability and composition of nutritional resources. Indeed, this would comport with expectations under life history theory and host-helminth co-evolution[56,57], where under resource constraints one should expect the suppression of non-existential phenotypes, i.e., of "luxury phenotypes"[58]. Indeed, antihelminthic immunity, unlike antiviral or antibacterial immunity, may be one such luxury phenotype. Under this hypothesis, investment in complete immunity to helminths (reliant on type 2 immune pathways) is expected when nutritional resources are abundant; conversely, under resource limitation, only inflammatory and type 1 immunity would be maintained. Vaccines that benefit from inflammatory and type 1 immunity would generate stronger vaccine-specific immune responses. This is consistent with, and might provide a coherent account of, our and other's studies reporting a pro-inflammatory effect of protein scarcity[59,60]. The elucidation of the mechanisms mediating these effects, specifically of the differential impact of nutrition on different immune pathways and functions, warrants further investigation.

Females overall responded better to vaccination than males, partly as a direct effect of sex (i.e. through unmeasured mediation) and partly mediated by body mass and fat content. Sexual dimorphism in disease and immunity is widely reported across multiple host and pathogen species, with females typically more resistant to infection including by helminths, generating stronger responses to vaccination, but more likely to develop autoimmunity[8,61–64]. Large-scale studies have identified sexual dimorphism in specific immune pathways, with females tending to present higher activation of innate pathways[65], though this might be especially true in younger individuals[66]. In our study, mice tended to be in the early adult stage, possibly skewing younger in the wild cohort, though this effect was vastly overshadowed by other negative effects on vaccine responsiveness. When adjusting for habitat, while females had lower body mass and fat stores, these differences had no effect on DTV-specific antibody responses. In fact, we only detected a direct effect of sex, and no indirect effects.

Importantly, our study was limited to variables easily measured in a wild, non-model, species. Further studies are needed to disentangle how different arms of the immune system mediate the effects of vaccination on antibody production, and how the environmental factors may modulate those different pathways. We also focussed on a single antibody isotype, IgG1. While this is appropriate for assessing alum-adjuvanted vaccines in a neutralising antibody scenario, further studies should investigate protective immunity to specific pathogens. Indeed, while our conclusions are consistent with reports that investigate protective immunity[18], it is likely that inferences would differ depending on the pathogens targeted. Aside from the immune system, it might be useful to explore how the microbiome might mediate the dietary effects, and how different nutrients, in quality and abundance whether supplied or available in the environment, might drive these interactions.

Lastly, taken together, this study provides a useful methodological approach to estimating causal relationships between environmental and individual factors and vaccine responsiveness. This approach could be used to predict vaccine responsiveness in target populations, and to identify which populations or subsets are likely to respond poorly to a specific vaccine, and benefit from targeted interventions such as diet changes, antiparasitic treatments, or different vaccine adjuvants. This could be particularly useful for vaccines that are known to be sensitive to environmental factors, such as anthelmintic vaccines, and for which the causes of vaccine hyporesponsiveness remain difficult to identify, quantify, and therefore, address. Our study also indicates that laboratory studies may not always be generalisable to wild populations, and are likely to systematically overestimate vaccine efficacy.This suggests that the development of vaccines should be more closely aligned with the conditions in which they are intended to be used, and that the effects of environmental factors on vaccine responsiveness should be more thoroughly investigated.

MATERIALS, METHODS, AND MODELS

All animal work was conducted in accordance to UK Home Office in compliance with the Animals (Scientific Procedures) Act, 1986. Both laboratory and field experiments were approved by the University of Edinburgh Ethical Review Committee and carried out under the UK Home Office Project Licence 70/8543. The dosage of diphtheria vaccine was tested in laboratory-bred wood mice before the experiments for safety and efficacy. Animal sacrifice was performed using appropriate Schedule 1 Methods. Fieldwork was carried out with permission of the Forestry Commission Scotland under the permit SUR09.

Laboratory wood mouse experiment

The laboratory experiment was carried out on a colony of wild-derived but now captive outbred wood mouse colony that has been bred and maintained at the University of Edinburgh[35] (See SI for more details). The experiment was carried out in two replicate blocks, with 36 animals in each block (18 males, 18 females). At 12 days before the start of the experiment (d-12), all mice were shifted to new diet regimes and given time to acclimatise; half of the animals in each block (9 males and 9 females) were given TransBreed™, a high quality whole-diet mouse food. We have previously shown that wood mice supplemented with TransBreed™ were more resistant to the gastrointestinal nematode *Heligmosomoides polygyrus*, cleared worms more effectively after anthelmintic treatment and produced stronger general (total IgA) and parasite-specific (IgG) immune responses in both wild and laboratory conditions[35]. The other half of the mice in this experiment were fed RM1 (Rat and Mouse Maintenance), a lower quality diet, but commonly used maintenance mouse food that supports maintenance. Both TransBreed™ and RM1 are standard laboratory diets, with TransBreed™ containing higher amounts of starch, protein, fat, and micronutrients compared to RM1 (see SI for detailed nutritional compositions). Within each diet regime, both male and female animals were randomly assigned to three experimental groups (Fig. 1). Animals belonging to Groups 1 & 2 (n = 24) were subcutaneously injected with 100µl of diphtheria vaccine (2 Lf per 100µl of adjuvant (9% Potassium Aluminium Sulphate) on d0. The vaccine was prepared in our laboratory one day before administration, using vaccine-grade diphtheria inactivated toxoid protein (Alpha Diagnostic International) adsorbed to the adjuvant (9% Alum). Animals assigned to Group 3 (n = 12) were injected with a control, referred to as 'adjuvant'; 100µl of adjuvant

(9% potassium aluminium sulphate) suspended in 1X PBS on d0. Small volume blood samples were taken via tail snip 14 and 17 days after the first injections (d14 and d17) to measure the primary antibody response. Twenty-one days (d21) after the first injections, the animals of Group 1 (n = 12) were injected again with 100µl of the diphtheria vaccine (referred to as the 'booster'), while those assigned to Group 2 (n= 12) were injected with the control dose as described above (referred to as 'adjuvant'). Animals assigned to Group 3 (n = 12) that had previously been injected with the control dose now received an injection of 100µl diphtheria vaccine. Another blood sample (20- 50µl) was taken a day later (d22) by cheek venipuncture. All animals were sacrificed 14 days after the second injection (d35) and a blood sample for measuring the secondary antibody response was taken. The same procedures and timeline were carried out for the second experimental block of animals (n = 36). Mice were co-housed throughout the experiment, in same-sex groups of three, with equal representation of the three experimental groups in each cage.

Wild wood mice experiment

We conducted a 10-week field experiment in a natural population of wood mice with a design similar to that of the laboratory experiment. The experiment took place in a woodland in Falkirk, Scotland, UK (Callendar Wood, 55.990470, -3.766636), where we established four trapping grids (60m × 40m per grid) with 10 m spacing between each of the 35 trapping stations. At each station, we set a pair of Sherman live traps (H.B. Sherman 2X2.5X6.5-inch folding trap, Tallahassee, FL, USA). We randomly selected two of the four grids to be 'supplemented'; 12 days before we began live-trapping, we started supplementing these grids with extra resources. Specifically, we scattered 6 kg TransBreed^{TM} (high quality, whole- diet mouse food) pellets evenly around the grid, and this supplementation continued throughout the experiment, twice per week. The two 'control' grids did not receive any supplementation. Wood mice in all four grids had access to their normal diet and food items, so these additional resources on the supplemented grids served as an additional high nutritional food resource.

From July-September 2018, wood mice were trapped 2–3 nights per week using live traps baited with grains, carrots, and bedding. Additionally, traps on the supplemented grids were also baited with 1-2 TransBreed^{TM} pellets. For identification, all newly captured mice weighing >13g were subcutaneously tagged with a unique 9-digit microchip passive induced transponder (PIT tag; FriendChip AVID2028, Norco, CA, USA). On first capture, mice were assigned to one of the three vaccine treatment groups described above in the laboratory experiment; mice assigned to these groups received the same primary and booster doses of the diphtheria vaccine or control injections as described below (see Figure 1). Throughout the experiment, weekly small volume blood samples were taken via tail snip from each tagged animal to measure their antibody response to the diphtheria immunisation. Animals recaptured 14 or more days after their second injection were sacrificed and a terminal bleed was collected to measure their antibody response and adult *Heligmosomoides polygyrus* worm burdens. *H. polygyrus* is a natural gastrointestinal nematode found at high prevalence (20-100 %) in wild populations of *A. sylvaticus*[35,67–69], as well as being a well-studied model system of human gastrointestinal nematodes where it has been found to be highly immunomodulatory[51,70].

Additionally, at all captures, we took the following demographic metrics for each individual: sex, body condition, reproductive condition, weight (g), and length (mm). Sex and reproductive condition of each mouse was assigned by examining the genitals, with males having larger urogenital gap compared to females. Males were classified as reproductively active if their testes had visibly descended to scrotal sacs, and females were classified as reproductively active if they had a perforated vagina or were visibly pregnant or lactating. Body condition was measured by assigning dorsal and pelvic fat scores on a qualitative scale from 1-5, where 1 represented a mouse with very low condition, while a 5 represented a mouse with ample fat reserves. This was achieved by palpating the back and pubic bones[71].

Vaccine responsiveness

Vaccine responsiveness was measured as the concentration of anti-diphtheria IgG1 antibodies; this is the gold standard method used in vaccinology to measure the antibody-specific response to immunisation and is used specifically for the diphtheria toxoid vaccine used here[72]. To measure diphtheria vaccine responsiveness, we collected blood samples from both the laboratory and the wild mice and these were centrifuged (@24,000 rpm for 10 min) within 4–6 hours of collection. The sera were separated from the pellets and stored at -80 °C until further analysis. We used ELISAs adapted for diphtheria-specific antibodies from Clerc et al. (2019a), to quantify anti-diphtheria IgG1 antibodies in the sera samples. 96-well plates (NuncTM MicroWellTM) were coated overnight at 4 °C with diphtheria toxin (2μg/mL) diluted in carbonate buffer (50 μl per well). After washing the plates with TBS (10 X) and Tween80 thrice, 100 μl of TBS(1X)-4percent BSA was added per well and incubated at 37 °C for 2 hours to block non-specific binding sites. Plates were then tapped dry and 50 μl of serum samples serially diluted (starting dilution 1:100) in TBS (1X)-4percent BSA buffer were added per well and left at 4 °C for binding overnight. The next day, 50 μl HRP-conjugated anti-mouse IgG1 detection antibody (Southern BioTech) per well was added after washing the plates 4 times with TBS (10X) and Tween80 and tapping them dry. After incubation at 37 °C for 1 hour, the plates were washed again, first 4 times with TBS (10X) and Tween80 and then twice with dH20. Then, 50 μl of TMB substrate solution were added per well and the enzymatic reaction was left to develop in the dark for 7 min. The reaction was stopped after 7 min using 50 μl sulphuric acid (0.18 M) per well. Absorbance at 450 nm was recorded using an ELISA plate reader (Multiscan, Ascent Labsystems) immediately thereafter. For each sample, blank-centred optical densities (OD) of three consecutive wells (of dilutions 1:3200, 1:6400, 1:12800) were averaged to obtain diphtheria-specific IgG antibody concentration.

Data processing and analyses

ELISA optical densities of DT-specific antibodies (E) were $1 + log_{10}$-transformed and, body weight (M), fat scores (F) and days post-vaccination ($T \geq 1$), were Z score-transformed to help the selection of sensible priors and to improve computational stability. All models were adjusted for time T between vaccination and blood collection to account for natural kinetics of the antibody response. Fat scores ranged from 2 to 5 and were derived from the sum of two five-point scales for pelvic and dorsal fat, where 1 is 'bony' to palpation and 5 is 'fat', and were treated as continuous for the analysis. All other variables were treated as binary.

All data processing was performed in Julia v1.11[73] using packages CSV.jl[74] and DataFrames.jl[75].

Generalised linear mixed models of the effects of experimental interventions on habitat, diet, and vaccine formulation

Generalised linear mixed models (GLMM) were used to estimate the effects of habitat, diet, and vaccine formulation on vaccine-specific antibody concentrations measured as optical density read-outs. Mouse ID was modelled as a random effect to account for multiple testing. The interactions between vaccine regimen and habitat, and between diet and habitat, were included in the full model to test whether mice in the wild would respond differently to vaccine timing and boosting or to diet supplementation than laboratory mice. Likelihood ratio tests (LRT) were used to test the contribution of the interaction terms to the model. We used Julia v1.9[73] package MixedModels.jl v4.13[76] was used for GLMMs and LRTs.

We used Bayesian multilevel models to minimise the effects of imbalance on our analyses and to quantify the uncertainty of those estimates. The intercept and slope priors were chosen to conform to the observed data, with equal means and 2.5 × the observed variance (see Supplementary Data: Multilevel Models). Typically, intercepts were given Gaussian priors of mean $\mu = mean(y)$ and 2.5 × σ those of the observed response variable y while population-level coefficients were given multivariate normal (Gaussian) priors of mean $\mu = 0$ and variance $\sigma = 2$. Population-level residuals were assumed to follow an exponential distribution with scale $\theta = 1/std(y)$. Parameters for group level intercepts and coefficients were given Gaussian priors and their variances were given positive-truncated Cauchy priors with location $x_0 = 0$ and scale $\gamma = 2$. Posterior estimates were sampled using the Hamiltonian Monte Carlo algorithm No U-Turn Sampler[77], typically across 4 threads simultaneously with $> 3,000$ draws. Turing.jl[78] was used for Bayesian modelling; Makie.jl v0.17[79] was used for plotting.

Structural Causal Models

The structural causal model (SCM) framework[36,80] proposes explicit qualitative solutions to biases inherent in datasets and can guide the statistical estimation of the mechanisms by which data are generated in complex systems[81,82]. Further, Bayesian statistical methods for estimating the causal effects indicated by the SCM allow flexibility of unbalanced small data and missing values[83], and provide explicit estimation of uncertainty given assumptions made by the model[84]. Specifically, here we have made three important assumptions: vaccine, diet, and habitat treatments were randomly allocated; there were no unmeasured (statistically relevant) confounders; and all relationships between variables were linear.

Model construction

Each of the variables X described above was included in the structural causal model $\mathbb{C}_{VE}$ with the following assignments of the form $X_j := f(X_{k \neq j}) + \mathcal{U}_j$ where the unmeasured noise variable $\mathcal{U}_j \sim Normal(0, 4)$, was assumed to be independent and identically distributed, and $f()$ was assumed to be linear (Supplementary Data: SCM construction):

A directed acyclic graph (DAG) expressing $\mathbb{C}_{VE}$ (Model 1) was used to build unconfounded models to estimate causal effects between each variable (Figure 3).

Model construction was performed in Julia v1.11[73] using package Dagitty.jl and Turing.jl[78], and $\mathbb{G}_{VE}$ corresponding to $\mathbb{C}_{VE}$ was drawn in Tikz[85].

Model validation

To validate $\mathbb{C}_{VE}$ (Model 1), we tested the independencies it implies following chain, fork, and collider rules[36,86]. Specifically, we used the observed values for each of the nodes of the causal graph (Figure 3) to test the truth of the following 13 implied independences: $(D \perp\!\!\!\perp S)$, $(D \perp\!\!\!\perp T \mid H)$, $(D \perp\!\!\!\perp V)$, $(F \perp\!\!\!\perp V \mid T, H)$, $(H \perp\!\!\!\perp S)$, $(H \perp\!\!\!\perp V)$, $(M \perp\!\!\!\perp V \mid T, H)$, $(P \perp\!\!\!\perp T \mid D, R, S, H)$, $(P \perp\!\!\!\perp V \mid T, H)$, $(P \perp\!\!\!\perp V \mid D, R, S, H)$, $(R \perp\!\!\!\perp V \mid T, H)$, $(S \perp\!\!\!\perp T)$, and $(S \perp\!\!\!\perp V)$. For each of these tests, we constructed a generalised linear mixed model with the appropriate link function, with mouse ID and immunisation protocol as random effects, and considered the statement true if $P_X > 0.1$, where $X \subseteq \mathbb{C}_{VE}$ (Supplementary Data: SCM construction). We used MixedModels.jl v4.13[76] to test the conditional independencies implied by $\mathbb{C}_{VE}$.

Model parameterisation

Next, we encoded $\mathbb{C}_{VE}$ as a Bayesian network in Turing.jl[78] to represent each variable as a probability distribution to be updated by the observed values. This allowed us to capture conditional expected values for each variable and their uncertainty. Models were constructed as described in Generalised linear mixed models of the effects of experimental interventions on habitat, diet, and vaccine formulation, code provided in Supplementary Data: SCM specification. Posterior estimates were validated with MixedModels.jl[76].

Missing data. Twenty-two mice had no fat scores, resulting in 67 missing values in the full repeated sampling dataset. To avoid having to remove those mice from the analysis, we used Bayesian imputation to estimate plausible probability distributions of each of the missing values in light of its parents (PA) in $\mathbb{C}_{VE}$ (Model 1), where $F := f(PA_F) = f(T, H, D, P, S)$ (Figure 3).

$\dot{F}$ denotes the variable F that contains imputed values; E_i denotes the expected distribution of E given $\dot{F}$.

Model updating and sampling. All Bayesian modelling was performed in Julia v1.9[73] using package Turing.jl v0.25[78].

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COMPETING INTERESTS

The authors declare that there are no competing interests.

AUTHOR CONTRIBUTIONS

Conceived and designed the experiments: ABP, SAB

Performed the experiments: SV, AS, ABP, SAB

Analysed the data: SV, SAB

Contributed reagents/materials/analysis tools: ABP, SAB

All the authors reviewed the manuscript.

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TABLES

Table 1: Posterior predictions of model parameters for random intercept Gaussian model $E \sim Normal(\alpha[ID] + H + D + V + V : H, \sigma)$.

Parameter	Mean	Std	5%	50%	95%
Intercept	1.02	0.85	-0.39	1.032	2.43
Habitat	-0.519	0.806	-1.863	-0.513	0.795
Diet	-0.253	0.072	-0.373	-0.253	-0.137
A	0.0	0.871	-3.307	-1.892	-0.438
D	2.062	0.856	-1.228	0.175	1.598
AD	1.256	0.875	-2.061	-0.651	0.84
DA	3.652	0.87	0.343	1.751	3.215
DD	2.996	0.865	-0.304	1.091	2.534
A:H	0.0	0.818	-0.864	0.47	1.831
D:H	-0.594	0.811	-1.434	-0.125	1.223
AD:H	-0.371	0.823	-1.244	0.103	1.472
DA:H	-1.348	0.818	-2.228	-0.874	0.496
DD:H	-0.527	0.818	-1.389	-0.061	1.31

Environmental drivers of low vaccine responsiveness in a lab-to-wild rodent model

Simon A. Babayan[1,†,*], Saudamini Venkatesan[2,†], Jessica Hall[1,2], Ewan Smith[1], Amy Sweeny[2], and Amy B. Pedersen[2,*]

[1]School of Biodiversity, One Health & Veterinary Medicine, University of Glasgow, Glasgow, UK
[2]Institute of Ecology and Evolution, School of Biological Sciences, University of Edinburgh, Edinburgh, UK
[†]These authors contributed equally to this work
[*]Corresponding authors: simon.babayan@glasgow.ac.uk, amy.pedersen@ed.ac.uk

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SUPPLEMENTARY INFORMATION

Supp methods

Insert additional details about the lab mouse colony details of the diet make up of Trans-Breed and RM1

Supplementary Data: Multilevel Models

Supplementary Data: SCM construction

Supplementary Data: SCM specification

Supplementary Data: Wood mouse data

Supplementary Figures

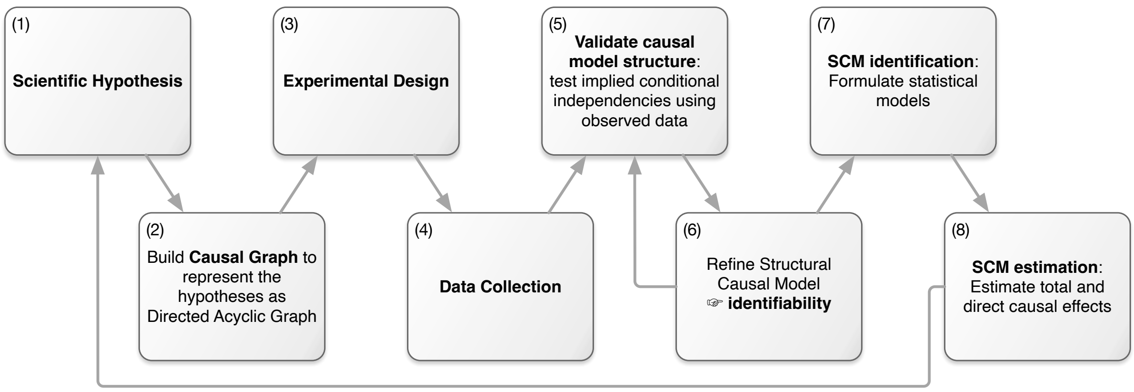

Fig. S1: Flow diagram of the SCM. The Structural Causal Model (SCM) process explicitly represents scientific hypotheses (1) as a directed acyclic graph (DAG). This DAG is mathematically encoded as a set of nested equations that describe the flow of causation between variables, and helps inform lab and field experimental design (3) and data collection (4). After processing, the data are used to test the validity of the causal assumptions underlying the SCM, e.g. by testing the conditional independencies implied by the DAG (5). If the assumptions are not supported, refinements of the SCM (6) are necessary. When all conditions are met, the SCM is identifiable and statistical models can be formulated to adjust for confounding (7). Parameters from these models can then be used as estimates of direct and indirect causal effects (8).

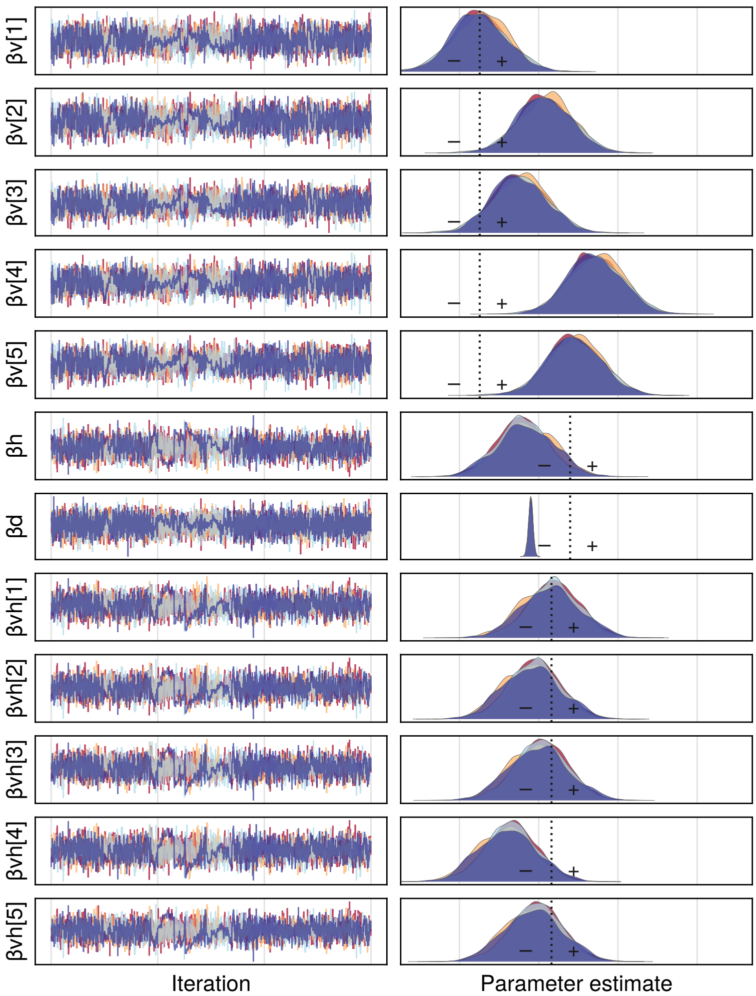

Fig. S2: MCMC chains and posterior distributions of coefficients for Diet, Habitat, Time post immunisation, and vaccine formulations A, AD, D, DA, and DD.